MICHAEL AHERNE

Leadership. Aerospace. Software.

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St. Louis, MO, USA
Dual US/Irish Citizen
Inactive Secret Clearance

EXPERIENCE

2024–PRESENT UMBRA — REMOTE

Director, Software Engineering

Umbra builds and operates a commercial SAR constellation and the software systems that plan missions, fly spacecraft, and deliver data products. I lead a 40-person ground software engineering organization spanning mission control, satellite scheduling, radar data processing, customer ordering and delivery, and reliability and platform engineering. Three managers covering seven teams report to me, along with three Staff-level individual contributors. Selected accomplishments:

- Built a repeatable hiring system adopted beyond my organization: authored job requisitions and a behavioral question bank, coached managers on assessment and interviewing, and migrated from Bamboo to Workable with Coderbyte, geo-mapping, cheating detection, and LLM-assisted resume scoring, cutting manual triage by roughly 80 percent.
- Built Staff-level Data and GroundOps capabilities by hiring Umbra's first Data Engineer at Staff level and two Staff GroundOps engineers, closing gaps in data, DevOps, and integration testing; coached managers through roughly 15 additional engineering hires.
- Co-authored and published a software-wide career ladder and management matrix, giving engineers and managers a clearer growth framework and a reusable foundation for scaled performance management.
- Led technical discovery and architectural direction for a next-generation mission-planning system.
- Bridged ground software, flight software, operations, and mission-facing teams, including loaning two engineers to a priority flight-software initiative and aligning mission-control and spacecraft-operations work.
- Advocated for improvements to spacecraft software architecture, deployment safety, distributed-system reliability, and onboard data processing, influencing technical roadmaps across flight and ground systems.

2022–2023 1904LABS — ST. LOUIS, MO

Senior Director, Practices & Solutions

1904labs builds software and data solutions for enterprise clients. I reported to the owner with responsibility for the company's technical practices and solutions. Selected accomplishments:

- Led approximately 70 engineers across Modern Software Engineering, Data Engineering, and Decision Science (applied data science), with three practice directors reporting to me and responsibility for delivery escalations below the owner.
- Set the strategic roadmap and guided the go-to-market strategy for AI-Enabled Customer Service, working with Sales and Marketing on positioning, partnerships, and enterprise pursuits.
- Coordinated the technical practices with Delivery, IT, Marketing, Business Development, and Human-Centered Design on solution development, demand generation, career paths, and market offerings.
- Mentored and coached three practice directors on leadership and provided career guidance to individual engineers.

EARLY 2022 **AHERNE, INC.** ST. LOUIS, MO

Father

Took a short break to support my new family. Selected accomplishments:

- Performed an estimated 3,000 successful diaper changes.
- Memorized classic literature through an extensive spaced-repetition training program, including Goodnight Gorilla, Little Blue Truck, and The Gruffalo.
- Led bottle prep and feeding operations in high-stress, time-critical situations.
- Oversaw a comprehensive, on-the-spot healthcare program specializing in kisses and ice packs.
- Sustained high morale with captivating airplane noises.

2021 **KUBOS CORP.** REMOTE

Director, Mission Control

Kubos makes a cloud-based satellite Mission Control Platform. As the 4th employee in the company, my responsibilities spanned software engineering, management, and support. Selected accomplishments:

- Spearheaded the establishment of the program vision and played a pivotal role in developing significant portions of the feature set. **[Ruby on Rails, React]**
- Significantly enhanced API scope and performance, resulting in improved efficiency and user experience. **[GraphQL, WebSockets, Python]**
- Assumed ownership of execution schedules and customer engagement, successfully coordinating the efforts of multiple Independent Contractors to ensure seamless project delivery.
- Provided comprehensive support to existing on-orbit customers, including round-the-clock Launch and Early Orbit Phase Support (LEOPS), guaranteeing the smooth operation of satellite missions.
- Generated documentation, how-to videos, blogs, and podcasts for both internal and external stakeholders.

2017–2021 **1904LABS** ST. LOUIS, MO

Director, Modern Software Engineering

1904labs helps leaders digitally transform their business by implementing modern software and data solutions. The Modern Software Engineering Practice contains all Full Stack, DevOps, and Mobile Engineers in the company. Selected accomplishments:

- Managed a personnel budget exceeding $5 million and a discretionary budget exceeding $40,000 for the Modern Software Engineering Practice.
- Created and sponsored a peer-led Study Group Program for technical learning, knowledge sharing, and mentorship across the practice.
- Designed a standardized Contribution Review process in which engineers assembled self-directed performance packets against a shared career framework.
- Led weekly Leadership Touchpoint meetings using prioritized agendas, minutes, and action tracking to coordinate work across the practice.
- Conducted screenings, interviews, and made hiring decisions for the Practice.

Senior DevOps Engineer

Embedded with three data engineers on Unified Transactions, a high-volume, low-latency project for a global payment network. Selected accomplishments:

- Learned Scala and implemented recursive depth-first-search matching in Spark and HBase to unify out-of-order payment events at a measured peak of 100,000 transactions per second, significantly accelerating transaction and chargeback investigations for banks. **[Scala, Spark, HBase]**
- Identified an organization-wide test-data gap and built a horizontally parallel Python data generator that produced billions of production-realistic events, exposing performance bottlenecks and helping establish production readiness. **[Python, Spark, HBase, ISO 8583]**

2015–2017 SAUCE LABS — BERLIN, GERMANY

Senior Software Engineer

Sauce provides a cloud-based platform for the automated testing of web and mobile applications. The scope of my responsibilities covered all aspects of the continuous integration pipeline. Selected accomplishments:

- Spearheaded multiple significant evolutions in the development process, introducing game-changing features such as automated branch testing, custom Slack integrations, a robust "follow the sun" PagerDuty setup, and Gated Commits. These advancements resulted in substantial time savings for developers. **[Jenkins, Python, CoffeeScript, Groovy]**
- Achieved a remarkable cost reduction of $120,000 per year by successfully re-architecting the data storage layer. **[AWS S3, CloudWatch]**
- Expertly managed a small team while also serving as an Agile Scrum Master, facilitating efficient and productive project execution and ensuring seamless collaboration within the team.
- Demonstrated exceptional skills in designing and building various web components to enhance the functionality and user experience of the Sauce platform. **[Angular, JavaScript]**
- Demonstrated a commitment to continuous learning by excelling in continuing education classes focused on machine learning, gaining proficiency in popular frameworks. **[TensorFlow, Octave]**
- Mentored junior engineers, provided valuable guidance and support, and received recognition for outstanding documentation practices, contributing to knowledge sharing and team effectiveness.

2012–2015 PLANET LABS — SAN FRANCISCO, CA

Director, Mission Control

Planet Labs is a multibillion-dollar Earth imaging company. As the 7th employee, I wore many hats, moving from the Spacecraft team to Manufacturing to Mission Operations. All told, I had an active role in designing, building, testing, and flying the first 113 satellites. Selected accomplishments:

- Designed and developed the majority of the microcontroller code for the first spacecraft, handling critical functions such as power management, inter-processor communication, scheduling, sensor acquisition, telemetry, and commands. Also contributed significantly to the codebase for the second spacecraft. **[C on PIC, then C on ARM]**
- Implemented the camera software responsible for capturing the first 10,000 photos. **[C++ on SBC]**
- Established and maintained the company's continuous integration and deployment system, starting with Vagrant and shell scripts, evolving to Jenkins and Ansible on OpenStack, and ultimately deploying Jenkins and Ansible on AWS. **[Vagrant, Jenkins, Ansible, OpenStack, AWS]**
- Co-initiated and maintained the company's code review system. **[Redmine, Phabricator]**
- Promoted to lead the Mission Operations team during the Flock 1a project, overseeing critical operations.
- Architected and programmed substantial components of the Mission Control system. **[Python/Django on Postgres, with monitoring (Nagios, New Relic, and Elasticsearch), satellite tasking (Celery and RabbitMQ), caching (Memcache and Redis), and user interfaces (JavaScript, jQuery, D3, Highcharts, Graphite, Bootstrap, and Backbone)]**
- Played a pivotal role in driving the long-term strategy for team composition. Conducted performance reviews, interviews, and managed the employee lifecycle before the company established an HR department.
- Led the expansion of remote worker infrastructure by advocating for the adoption of ChatOps. **[HipChat/CoffeeScript]**
- Co-led software development efforts for the Manufacturing and Production teams. **[REST API in Python/Flask, website in Python/Django, GSE in Arduino/Raspberry Pi]**
- Introduced frontend unit testing to the Manufacturing Team's continuous integration process. **[Backbone/Jasmine]**
- Mentored numerous interns and new hires across multiple teams, providing guidance and support to foster their professional growth.
- Gave tours to distinguished guests and actively shaped company culture, advocating for the unique artist-in-residence program.

2009–2012 **INFORMATION SCIENCES INSTITUTE** MARINA DEL REY, CA

Research Satellite Engineer

Designed guidance, navigation and control systems and managed students on several microsatellite research programs. Selected accomplishments:

- Solely programmed the entire flight software system for USC's first CubeSat, launched December 8, 2010 aboard SpaceX's Falcon 9 rocket. (See publication 1.) **[C on PIC]**
- Implemented the attitude control system for the first-ever surface-tracking CubeSat, launched in July 2012. (See publication 2.) **[C on PIC, MatrixX/Simulink on Windows]**
- Published research on rendezvous and proximity operations using a vision-based autonomous tracking system. (See publication 3.) **[OpenCV and Haar classifiers]**
- Refactored Mission Control as a website, enhancing functionality and user experience. **[PHP/MySQL]**
- Designed and programmed the control systems for thruster-based microsatellite prototypes, involving Kalman filtering, computer-assisted docking and PID and phase plane controllers. **[C on Rabbit]**
- Developed functional and environmental test requirements for the Aeneas CubeSat program and served as Integration and Test Director.
- Created an Application Programming Interface (API) for commanding microsatellites over a wireless TCP/IP network using a Rabbit 4000 microcontroller. **[C on Rabbit]**
- Managed a team consisting of both graduate and undergraduate students, providing guidance and supervision throughout their involvement in various projects.
- Designed the SERC website and took responsibility for creating and maintaining visual content, including pictures and videos. **[Photoshop, MS Movie Maker, Paint.NET]**

2006–2009 **STINGER GHAFFARIAN TECHNOLOGIES (SGT)** UPPER MARLBORO, MD

Systems Engineer

Managed NASA Goddard Space Flight Center's Requirements Database and designed satellite propulsion subsystems. Selected accomplishments:

- Created automated tools to help with propulsion subsystem design, including tank sizing, plume impingement considerations, delta-v budgeting and cost/weight estimations. **[Visual Basic]**
- Implemented significant cost savings through autonomous linking of requirements, and developed tools for parsing, characterization and trace development. **[Visual Basic]**
- Telecommuted for 2 years, receiving high praise on deliverables and performance evaluations.

2003–2006 **FAA OFFICE OF COMMERCIAL SPACE TRANSPORTATION** WASHINGTON, DC

Aerospace Engineer

Oversaw amateur rocket launches and regulation development. Conducted safety analyses for license and permit applications. Performed on-site safety inspections and compliance monitoring. Selected accomplishments:

- Maintained flawless safety record of the amateur rocket community.
- Drove final team concurrence on regulations that had been stalled for 12 years. (See publication 4.)
- Cut $100,000 in costs through prudent contract management.
- Elected by colleagues as Employee of the Year, elected by management as Top Performer.
- Experienced with all Range Safety analyses, including
  - 6-degree-of-freedom trajectory simulation, dispersion and malfunction turn analyses
  - Blast overpressure calculations, damage modeling, debris generation and fragmentation distance
  - Probability-of-impact and cumulative risk assessment
  - FMECA, Fault Tree Analyses, Hazard Analyses **[POST, OrSAT, STK, Splash, TaOS, Maple, Visual Basic]**

EDUCATION

COURSERA | UDACITY | UDEMY ONLINE

Continuing Education

- iOS & Swift - Dr. Angela Yu - Udemy
- Machine Learning - Andrew Ng - Stanford University
- Conflict Management - Najla DeBow - UC Irvine
- Software Testing - John Regehr - Udacity

2009–2012 **UNIVERSITY OF SOUTHERN CALIFORNIA (USC)** LOS ANGELES, CA

M.S. Astronautical Engineering

- Interdisciplinary coursework in control systems, robotics, filtering, estimation, mobile robot architectures and artificial intelligence.
- Noteworthy academic projects:
  - Path planning program (using A* search) for course in Artificial Intelligence
  - Sound localization using 3 microphones and LabVIEW for “Sensing and Planning in Robotics”
  - GNC engineer for LEAPFROG project – a hovering jet-based lunar-lander. Flew March 2009.

1999–2003 **EMBRY-RIDDLE AERONAUTICAL UNIVERSITY (ERAU)** DAYTONA BEACH, FL

B.S. Engineering Physics, Minor Mathematics

- Senior Design team placed 1st and 3rd in two separate NASA RASC-AL competitions.
- Junior design team competed in NASA’s 9th annual Great Moonbuggy Race.
- Recipient of the “Most Outstanding Student” Award.
- First engineering student to study abroad in Australia.

PUBLICATIONS

CAERUS – CONCEPT THROUGH FLIGHT IN ELEVEN MONTHS: A STORY OF RAPID RESPONSE AND LESSONS LEARNED.

J. Tim Barrett, Michael Aherne, Will Bezouska, Jeff Sachs and Lucy Hoag AIAA-2011-713. Presented at the 2011 AIAA Space Conference, Pasadena, CA.

COLONY I MEETS THREE-AXIS POINTING.

M. Aherne, T. Barrett, L. Hoag, E. Teegarden, R. Ramadas SSC11-XII-7. 2011 Utah Small Satellite Conference.

DEMONSTRATION OF TECHNOLOGIES FOR AUTONOMOUS MICRO-SATELLITE ASSEMBLY.

M. Aherne, T. Barrett, W. Bezouska and S. Schultz AIAA-2009-6504. Presented at the 2009 AIAA Space Conference. Pasadena, CA.

REQUIREMENTS FOR AMATEUR ROCKET ACTIVITIES.

Federal Aviation Administration. RIN 2120–2120–AI88. Federal Register, Vol. 73, No. 234, December 4, 2008.

LANGUAGES

English (C2, Native), German (A2, Waystage)

PERSONAL

Flying, Rollerblading, SCUBA, Snowboard, Drums, Hiking, Photography